**MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS**
**FALL SEMESTER 2026**
**TEST 1 (PRACTICE)**

Family Name: _____

Given Name: _____

UID: _____

**Instructions:**

(1) The question paper is on the **last page**. You may tear off the last page, but you must return the question paper at the end of the test.
(2) Write your answers on the attached A4 paper. Work written on draft paper or on the question paper will NOT be graded.
(3) For Questions 7 and 8, show your steps or reasoning clearly to receive full credit, unless otherwise specified in the question. For Questions 1 to 6, write your answers in the blanks provided on the next page.

**Write your work in this page**

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| C | F | E | E | F | E |

7(a). Since

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Let Let $\epsilon > 0$ such that $e - \epsilon > 1$. There exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|\left(1 + \frac{1}{n}\right)^n - e| < \epsilon.$$

Hence,

$$-\epsilon < \left(1 + \frac{1}{n}\right)^n - e < \epsilon.$$

Thus,

$$e - \epsilon < \left(1 + \frac{1}{n}\right)^n$$

Hence, for any $n > N$,

$$(e - \epsilon)^n < \left(1 + \frac{1}{n}\right)^{n^2}.$$

Since $((e - \epsilon)^n)$ is unbounded,

$$(\left(1 + \frac{1}{n}\right)^{n^2})$$

is also unbounded. Thus, $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n^2}$ $\boxed{\text{does not exist}}$.

7(b). Let

$$x_n = \sin\left(\frac{n\pi}{7}\right) + \frac{1}{n}.$$

Note that

$$x_{7k} = \frac{1}{7k},$$

and so

$$\lim_{k\to\infty} x_{7k} = 0.$$

Moreover,

$$x_{14k+1} = \sin\left(\frac{\pi}{7}\right) + \frac{1}{7k+1}$$

and so

$$\lim_{k\to\infty} x_{7k+1} = \sin\left(\frac{\pi}{7}\right).$$

Since there are two subsequences of $(x_n)$ coverging to two different limits,

$$\lim_{n\to\infty} x_n \boxed{\text{ does not exist.}}$$

8. $\boxed{True.}$ Suppose $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. Then $\sum_{n=1}^{\infty} |a_n|$ is convergent. Hence,

$$\lim_{n\to\infty} |a_n| = 0.$$

Thus, there exists $N \in \mathbb{N}$ such that for any $n > N$,

$$|a_n| < 1$$

Hence, $|a_n|^2 < |a_n|$ for all $n > N$. Thus,

$$\sum_{n=N+1}^{\infty} a_n^2 = \sum_{n=N+1}^{\infty} |a_n|^2 < \sum_{n=N+1}^{\infty} |a_n|.$$

Since $\sum_{n=N+1}^{\infty} |a_n|$ is convergent, the comparison test implies that

$$\sum_{n=N+1}^{\infty} a_n^2$$

is also convergent. Hence,

$$\sum_{n=1}^{\infty} a_n^2$$

is convergent.

**Write your work in this page**

**Write your work in this page**

**Sample Practice Test Paper (Fall 2026)**
**Writings on this page will not be graded**

Name:
UID:

## Section A: Multiple-Choice Questions.

For Questions (1)–(6), choose the one best answer from options A–H below.

A. Only (I) is correct.
B. Only (II) is correct.
C. Only (III) is correct.
D. Only (I) and (II) are correct.
E. Only (I) and (III) are correct.
F. Only (II) and (III) are correct.
G. All are correct.
H. None of the above.

(1) Which of the following statements is/are correct?
   (I) The Bolzano–Weierstrass theorem states that any subsequence of a convergent sequence is convergent.
   (II) There exists a sequence of numbers in $[-2026, 2026]$ that has no convergent subsequence.
   (III) There exists a sequence of rational numbers that has no convergent subsequence.

(2) Which of the following statements is/are correct?
   (I) Any field is ordered.
   (II) Let $F$ be an ordered field. Then $0 < 1$.
   (III) Let $F$ be an ordered field. For any $0 \neq a \in F$, $0 < a^2$.

(3) Which of the following statements is/are correct?
   (I) If $S \subseteq \mathbb{R}$ is nonempty and bounded above, and $c > 0$, then
   $$\sup(cS) = c\sup(S),$$
   where $cS = \{cs : s \in S\}$.
   (II) If $S \subseteq \mathbb{R}$ is nonempty and bounded above, and $c < 0$, then
   $$\sup(cS) = c\sup(S).$$
   (III) If $S \subseteq \mathbb{R}$ is nonempty and bounded below, and $c < 0$, then
   $$\sup(cS) = c\inf(S).$$

(4) Which of the following statements is/are correct?

(I) If $(x_n)$ is bounded and $\lim y_n = 0$, then $\lim(x_n y_n) = 0$.

(II) If $\lim(x_n y_n) = 0$ and $\lim y_n = 0$, then $(x_n)$ must be bounded.

(III) If $\lim x_n = L$ and $x_n \geq 0$ for all $n \in \mathbb{N}$, then $L \geq 0$.

(5) Which of the following statements is/are correct?

(I) A sequence $(x_n)$ converges to $L \in \mathbb{R}$ if, there exists $N \in \mathbb{N}$ such that for every $\varepsilon > 0$ and for all $n \geq N$,

$$|x_n - L| < \varepsilon.$$

(II) Every convergent sequence is bounded.

(III) Every monotone convergent sequence is bounded.

(6) Which of the following statements is/are correct?

(I) The series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{n^3 + 1}$ is conditionally convergent.

(II) The series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ is divergent.

(III) Any rearrangement of an absolutely convergent series is convergent.

**Section B: Long-Answer Questions.**

(7) Determine whether the following limits exist. If a limit exists, find it. Explain your answers.

(a) $\displaystyle\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n^2}$.

(b) $\displaystyle\lim_{n\to\infty} \left(\sin\left(\frac{n\pi}{7}\right) + \frac{1}{n}\right)$.

(8) Let $(a_n)$ be a sequence of real numbers. Prove or disprove the following statement: If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\displaystyle\sum_{n=1}^{\infty} a_n^2$ also converges absolutely.

**Axioms for an ordered field $F$:**

(O1) For any $a, b \in F$, either $a \leq b$ or $b \leq a$.

(O2) For $a, b \in F$, if $a \leq b$ and $b \leq a$, then $a = b$.

(O3) For $a, b, c \in F$, if $a \leq b$ and $b \leq c$, then $a \leq c$.

(O4) For $a, b, c \in F$, if $a \leq b$, then $a + c \leq b + c$.

(O5) For $a, b, c \in F$, if $a \leq b$ and $c \geq 0$, then $ac \leq bc$.